



The Action Potential - Synaptic Transmission (31-36)

The Action Potential

- **Action potential:** electrical potential across neuronal membrane when neuron is firing
- Measured in volts(V)
- Few milliseconds
- Triggered by any stimulus
- Involves rapid ion channel opening/closing
- AKA spikes/nerve impulses/APs



<https://www.youtube.com/watch?v=UZhGjcuTDs>

Rising Phase

- Resting neuron potential: -60 to -80 mV
- Membrane potential *rises* in voltage
 - Positive ions enter neuron
 - Neuron's charge approaches charge outside (depolarization)
- Potential get to around +30 to +40 mV(peak)
- When rising phase starts, full action potential *has to finish*



Falling Phase

- Membrane potential decreases
 - Positive ions leave neuron
 - Repolarization/hyperpolarization
- continues to fall until -80 mV
 - undershoot/afterhyperpolarization
- Na-K pump works to reset voltage



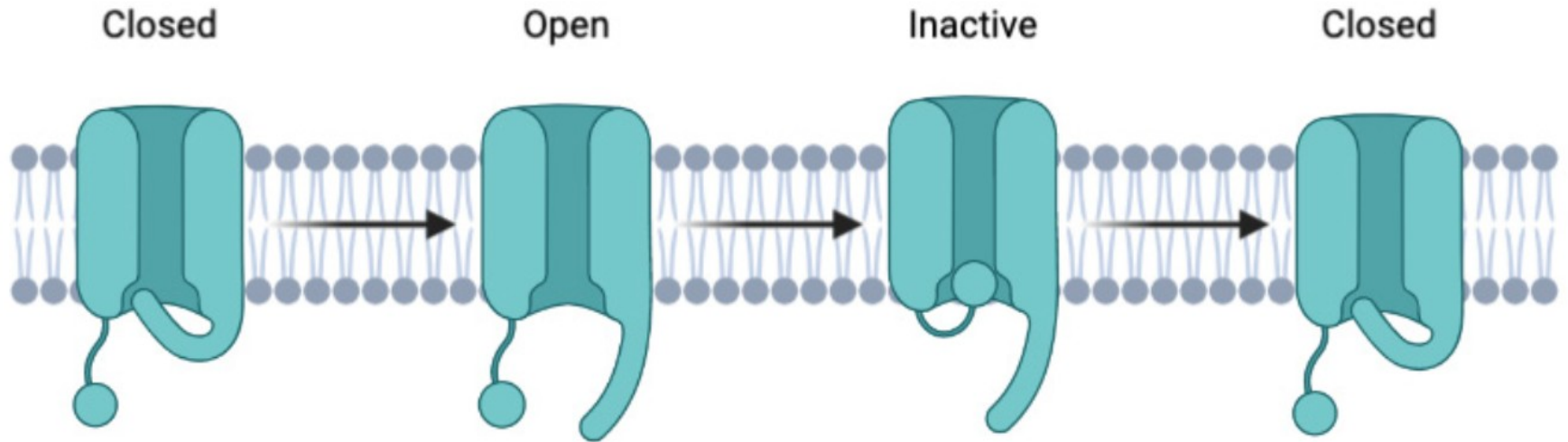
Sodium Channels

- Channels that let ions move at certain membrane potentials or voltages(Voltage-gated)
- Outside more positive than inside(Positive ions try to go inside)
- Rising phase due to Na^+ influx
- Peak(+30 to +40 mV), channels become inactive
 - Absolute refractory period prevents second AP



FIGURE 24

Voltage-Gated Sodium Channel States

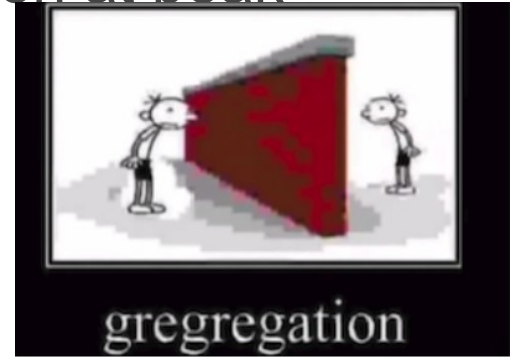


States of voltage-gated sodium channels.²⁵

Potassium Channels



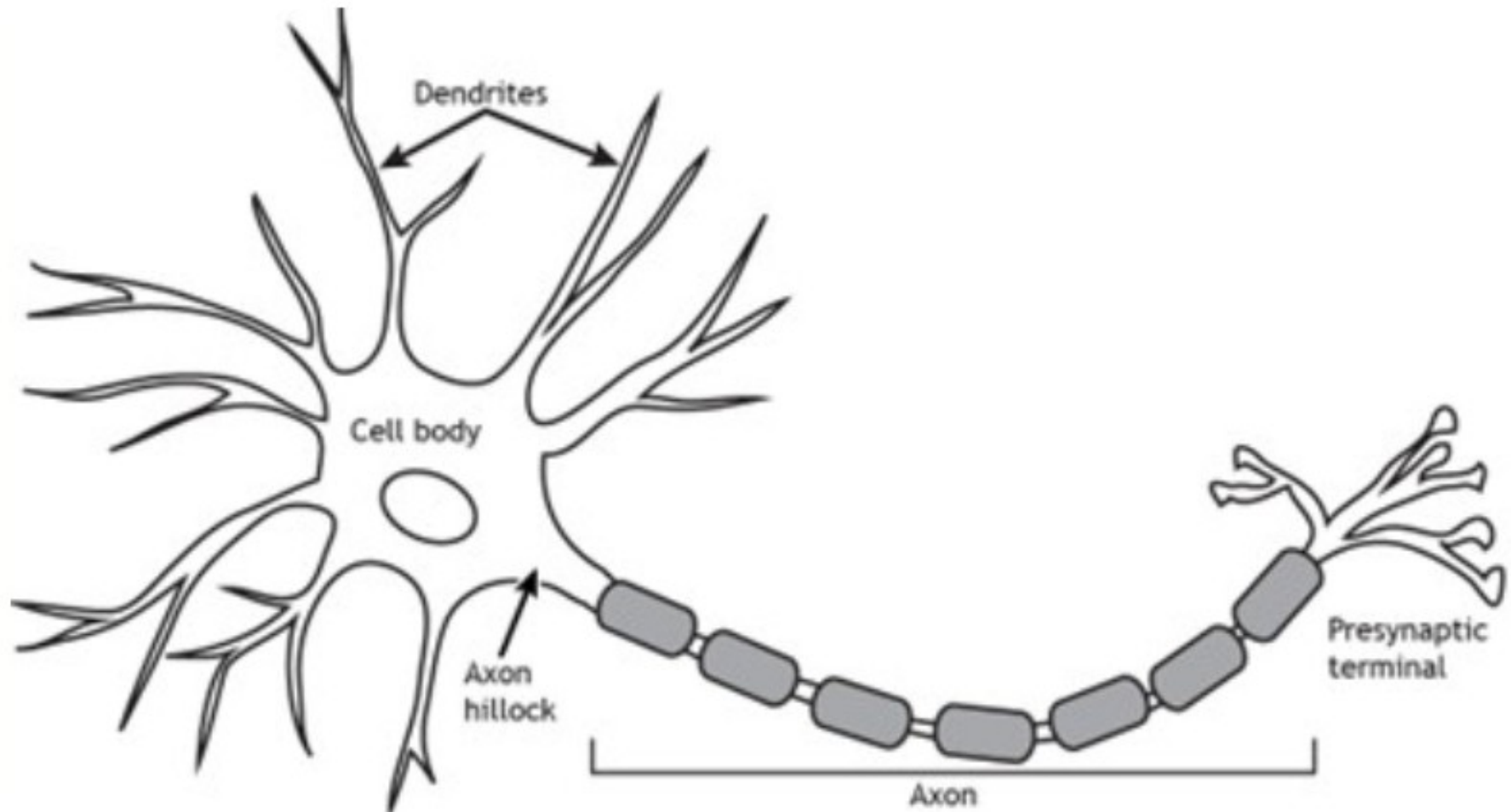
- Potassium channels start to open at 0 mV
 - Slow opening and closing, fully open at peak
- K^+ rushes out (K^+ Efflux)
- Start to close near resting potential
 - K keeps leaving
- Referred as delayed rectifier
- Still is impossible to start another AP(relative refractory period)
- Reset by with Sodium-potassium pump



Action Potential Conduction

- Initiated at the cell body(soma)
- Travels down Axon
 - Repeating the action potential sequence
 - Can trigger another sequence
 - Only goes in one way





The axon hillock is located where the cell body transitions into the axon and is the site where the action potential signal is initiated.²⁷

The Nodes of Ranvier and Saltatory Conduction

- Areas of axon is covered in myelin
 - Sheath made by glial cells
 - Similar to insulation on wires
- **Nodes of Ranvier:** gaps between Sheaths
- Na & K channels clumped in Nodes of Ranvier
- **Saltatory conduction:** conduction happening in myelin sheaths



I thank

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You!